

VerbalVis: Full-Duplex Conversational Visual Analytics for Analytical Intent Revision

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VerbalVis: Supporting Analytical Intent Evolution through Full-Duplex Conversational Visual Analytics

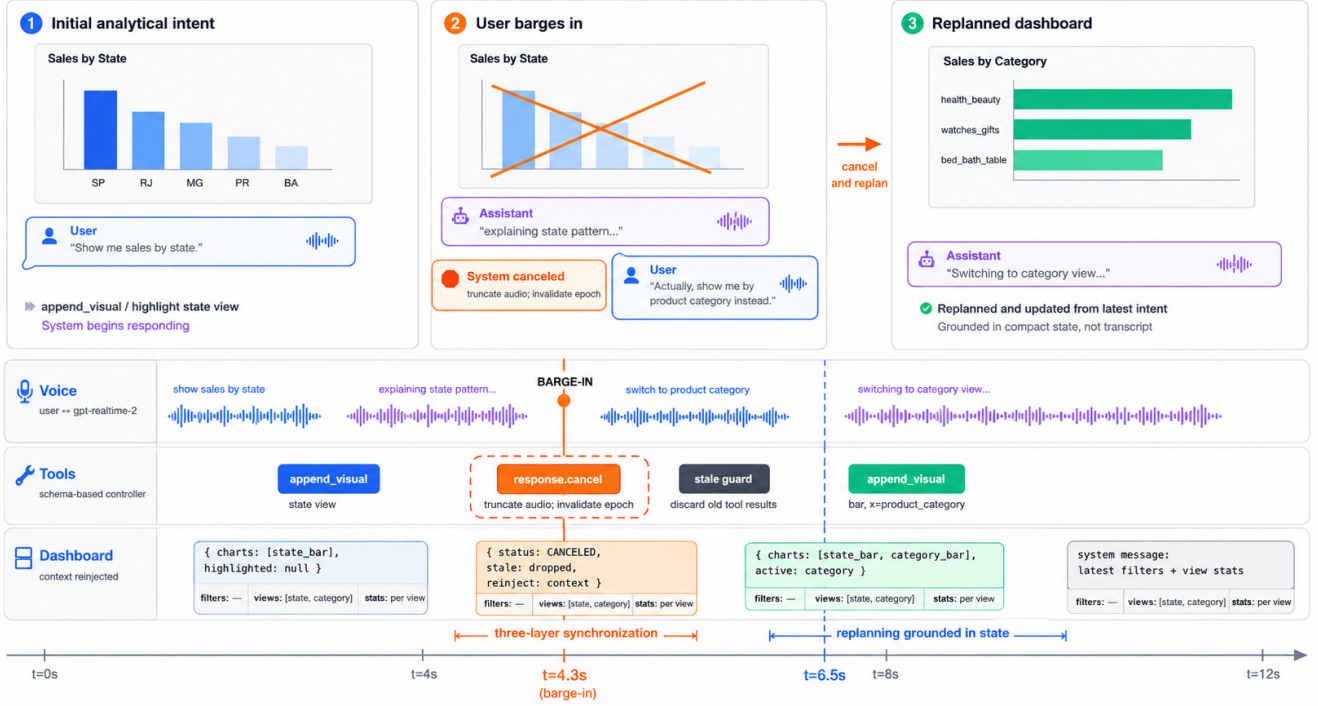


Fig. 1. Overview of a barge-in episode in VerbalVis. While the assistant is explaining a visualization, the user interrupts to redirect the exploratory analysis from sales by state to sales by product category. VerbalVis immediately cancels obsolete speech, replans visualization actions through tool execution, reinjects the updated dashboard state as contextual grounding for the language model, and synchronously updates both the dashboard and spoken response. The aligned voice, tool, and dashboard timelines illustrate the coordinated full-duplex interaction that enables continuous analytical intent evolution without explicit turn switching.

Abstract—Exploratory data analysis is iterative: observations may cause analysts to redirect their questions, revise provisional explanations, or change the data scope under investigation. Existing speech- and language-driven visual analytics systems generally support such revisions only after the current response has completed. Full-duplex speech interaction allows users to redirect an analysis while the assistant is still speaking, but stopping speech alone is insufficient when the superseded response may also have initiated analytical actions that modify a persistent dashboard. We present VerbalVis, a full-duplex conversational visual analytics system that coordinates spoken output, schema-grounded tool execution, and dashboard state during mid-response redirection. VerbalVis truncates obsolete speech, supersedes the active response, invalidates response-bound analytical actions, rejects stale results, and replans from the latest committed dashboard state. To describe analytical redirection, we distinguish three theory-informed, non-exhaustive, and potentially overlapping objects of revision: the analytical goal, a working hypothesis, and the analytical scope. These

constructs guide design inquiry and behavioral coding rather than runtime classification. A lightweight formative inquiry informed requirements for cross-layer cancellation, state-grounded replanning, and visible execution feedback. We evaluate VerbalVis through an illustrative scenario and an exploratory within-subjects study comparing full-duplex and turn-based conversational visual analysis.

Index Terms—Visual analytics, conversational interfaces, speech interaction, full-duplex dialogue, analytical intent revision, exploratory data analysis.

I. INTRODUCTION

EXPLORATORY data analysis (EDA) is inherently iterative. Analysts rarely follow a fully specified plan; instead, they move among observations, questions, tentative explanations, and new analytical actions as evidence emerges. Patterns and anomalies can expose limitations in the current direction and prompt analysts to reconsider what they are asking, what they believe, or which data are relevant [?], [?],

[?], [?]. We use *analytical intent revision* to denote an episode in which an analyst supersedes, qualifies, or materially changes an active analytical commitment.

Natural-language interfaces and LLM-powered visual analytics systems increasingly support contextual queries, visualization generation, conversational refinement, and multistep analytical tool use [?], [?], [?], [?], [?], [?]. These capabilities make it easier to express complex analytical requests and iteratively refine visual results. However, refinement is generally organized around completed turns: a user submits a request, the system produces an explanation or executes a sequence of analytical actions, and the user changes direction only after that response has finished. This organization is particularly restrictive for speech, where system output unfolds over time and an analyst may recognize a misdirected analysis well before the assistant has finished speaking. The interaction therefore delays the system’s response to a change that has already occurred in the analyst’s reasoning.

Full-duplex interaction provides a different regime by allowing the system to continue listening while speaking. An analyst can redirect an ongoing analysis through barge-in rather than waiting for a turn boundary. Yet barge-in alone does not solve the visual analytics problem. Barge-in is a temporal event, not necessarily evidence of analytical revision: a user may interrupt to correct recognition, request clarification, acknowledge the assistant, or stop an overly long response. More importantly, stopping speech does not necessarily stop the analytical consequences of the superseded response. Analytical tool actions and pending dashboard updates may continue after the user has changed direction, and late results may still modify the visible state. The assistant could therefore acknowledge the revised request while the dashboard continues evolving according to the obsolete one. Full-duplex conversational visual analytics consequently introduces a cross-layer consistency problem spanning spoken output, analytical execution, and persistent dashboard state.

We present **VerbalVis**, a full-duplex conversational visual analytics system designed to keep these layers aligned during mid-response redirection. VerbalVis combines real-time speech interaction, schema-grounded analytical tools, and a persistent dashboard through a visual-analytics-specific orchestration layer. When a new request supersedes an active response, the system truncates obsolete speech and advances a response-bound execution epoch. Tool actions remain associated with the response that produced them; actions and results from earlier epochs are treated as stale and cannot commit changes to the dashboard. The revised request is then interpreted against the latest committed dashboard state, including active filters, highlights, and visualizations, so that replanning begins from what the analyst currently sees rather than from speculative work associated with the cancelled response. VerbalVis does not introduce a new speech model or interruption detector. Its contribution is the response-supersession mechanism that coordinates speech cancellation, analytical validity, and dashboard-grounded replanning.

Drawing on research in sensemaking and exploratory analysis, we characterize analytical revisions through three potentially overlapping objects of change: the analytical goal,

a working hypothesis, and the analytical scope [?], [?], [?], [?]. These constructs provide a vocabulary for describing how an active investigation is redirected, but they do not imply that every interruption constitutes a revision or that every revision belongs to only one category. They serve as theory-informed sensitizing concepts and post-hoc coding dimensions rather than an exhaustive taxonomy or runtime intent classes. A lightweight formative inquiry refined their operational boundaries and informed requirements for cancellation, state preservation, visible feedback, and recoverable dashboard interaction.

We evaluate VerbalVis through an illustrative usage scenario and an exploratory within-subjects mixed-methods study with eight participants. The study compares a *Full-Duplex* condition, in which participants may redirect the assistant while it is speaking, with a matched *Turn-Based* condition, in which they must wait for the current response to finish. We examine interaction responsiveness, perceived control, state consistency, workload, exploratory behavior, and analytical outcomes, with particular attention to when participants revise an ongoing analysis and what they expect the system to cancel, preserve, and update.

Contributions.

- 1) A design framing for overlapping user speech in full-duplex conversational visual analytics. We distinguish non-interruptive backchannels from floor-taking barge-ins and characterize analytical revisions through changes to analytical goals, working hypotheses, and analytical scope.
- 2) **VerbalVis**, a full-duplex conversational visual analytics system that coordinates assistant speech, analytical tool execution, and persistent dashboard state when an active analytical response is superseded. It supports speech truncation, response-bound tool execution, stale-result rejection, and replanning from the latest committed dashboard state.
- 3) An exploratory mixed-methods comparison of full-duplex and turn-based speech interaction during open-ended visual analysis, examining responsiveness, perceived control, interaction flow, state understanding, workload, exploratory behavior, and analytical outcomes.

II. RELATED WORK

VerbalVis connects research on natural-language interaction for visualization, LLM-powered visual analytics, full-duplex spoken dialogue, and analytical sensemaking. We review how these areas support conversational refinement, analytical steering, interruption, and changes to an ongoing investigation.

A. Natural-Language Interaction for Visualization

Natural-language interfaces for visualization translate utterances into data attributes, analytical tasks, and visual specifications. DataTone exposes alternative interpretations through interactive ambiguity widgets [?], while Eviza grounds language in both the dataset and the active visualization to support contextual references and follow-up questions [?]. NL4DV

provides a reusable pipeline for extracting attributes and analytical tasks and generating candidate visualizations [?]. These systems establish language grounding, ambiguity management, and contextual interpretation as central requirements for conversational visual analysis.

Multimodal systems extend this interaction by combining language with visible references and direct manipulation. Orko integrates speech and touch for network exploration [?]; InChorus characterizes operations expressed through speech, pen, and touch [?]; and DataBreeze interweaves speech with direct manipulation over flexible unit visualizations [?]. Snowy and conversational extensions of NL4DV further support follow-up utterances and modification of existing views [?], [?]. However, these systems generally organize refinement around completed user and system turns. VerbalVis instead considers redirection while speech and associated analytical actions are still unfolding.

B. LLM-Powered and Agentic Visual Analytics

Large language models extend conversational visualization beyond mapping one utterance to one chart. LIDA decomposes visualization generation into data summarization, goal generation, chart creation, and evaluation [?]. Data Formulator combines natural-language concepts with directly specified visual encodings, while Data Formulator 2 supports iterative authoring and navigation through alternative data and visualization histories [?], [?]. These systems broaden the role of language from chart specification to data transformation and iterative visual design.

Agentic systems further delegate analytical planning and execution to LLM-based components. InsightPilot and LightVA translate higher-level analytical goals into sequences of tasks, visualizations, and findings [?], [?]. WaitGPT externalizes intermediate analysis operations for inspection and modification [?], while InsightLens links conversation, code, visualizations, and findings across an extended analysis [?]. Such systems make analytical processes more steerable and inspectable. Nevertheless, steering is usually performed through subsequent prompts, graphical controls, or explicit checkpoints. Existing work rarely specifies whether an unfinished action generated by an earlier response remains valid after the user has redirected the analysis.

C. Full-Duplex Dialogue and Interruption

Streaming speech systems incrementally process or generate audio, whereas full-duplex systems continue listening while speaking. They must distinguish background speech, backchannels, cooperative overlap, and attempts to take the conversational floor [?]. Barge-in has traditionally referred to detecting user speech during system output and stopping or modifying the active response [?]. Recent systems including Duplex Conversation, SyncLLM, Moshi, and SALMONN-omni investigate concurrent speech streams, low-latency turn-taking, and context-sensitive interruption [?], [?], [?], [?].

Interruption timing, however, does not determine analytical meaning. A user may interrupt to acknowledge the assistant,

repair recognition, request clarification, or revise the investigation. Conversely, an analyst may revise the analysis after the assistant has finished speaking. Moreover, stopping audio does not automatically cancel a database query, visualization tool call, or pending dashboard update. VerbalVis therefore treats barge-in as a temporal signal while separately managing the validity of response-dependent analytical work and persistent visual state.

D. Sensemaking and Analytical Revision

Sensemaking research characterizes analysis as an iterative process in which questions, explanations, and data focus evolve as evidence is encountered [?], [?], [?]. For *Analytical Goal Shift*, information-seeking studies describe evolving queries and the refinement of broad goals into focused analytical subtasks [?], [?], [?]. For *Working-Hypothesis Revision*, Data-Frame Theory and scientific reasoning show that provisional explanations may be questioned or replaced when observations conflict with prior expectations [?], [?], [?]. For *Analytical Scope Refinement*, information-foraging and visualization task frameworks distinguish analytical purpose from the data target and its progressive restriction by population, variable, time, geography, category, or granularity [?], [?], [?].

These traditions provide theoretical support for the three forms, but do not establish a universal, exhaustive, or mutually exclusive taxonomy. We therefore treat them as theory-informed sensitizing concepts. Section ?? presents participants’ utterances as empirical instances and specifies the coding boundaries adopted in this work; it does not serve as the theoretical evidence that these forms of revision exist.

Across these areas, prior work supports contextual language interaction, multistep analytical execution, spoken interruption, and evolving analytical histories. VerbalVis connects these capabilities by coordinating the validity of spoken output, analytical actions, and persistent dashboard state when an unfolding response is superseded.

III. FORMATIVE INQUIRY AND DESIGN REQUIREMENTS

We conducted a formative inquiry with four participants (F01–F04) to understand how analysts revise an ongoing investigation during conversational visual analysis and what system behavior such revisions require. The inquiry was design-oriented rather than confirmatory: it aimed to refine operational definitions, identify compound and boundary cases, and inform the design of VerbalVis, rather than establish an exhaustive taxonomy of analytical behavior.

A. Theory-Informed Revision Constructs

Exploratory analysis is iterative: observations can redirect questions, reshape provisional explanations, and change which data are considered relevant [?], [?], [?]. Research on sensemaking and scientific reasoning further shows that analysts construct and revise explanatory frames as they encounter new evidence [?], [?], [?]. Information-seeking and visualization-task research similarly distinguishes changes in analytical purpose from changes in the data target and means of analysis [?], [?], [?].

Based on these traditions, we specified three theory-informed sensitizing concepts before analyzing the formative sessions:

Analytical Goal Shift materially redirects the primary analytical question or desired knowledge outcome.

Working-Hypothesis Revision replaces, or qualifies an active provisional explanation, interpretation, or decision-relevant proposition.

Analytical Scope Refinement narrows or broadens constraints over the relevant population, time period, geography, variables, categories, granularity, or data subset while generally preserving the higher-level analytical objective.

These constructs describe different objects of analytical change rather than mutually exclusive event classes. A single utterance may therefore receive multiple labels. They are also non-exhaustive: hypothesis formation, method changes, representation changes, evidence requests, and conversational repair were retained as separate boundary codes.

B. Procedure and Analysis

Participants used an early VerbalVis prototype to explore the Olist Brazilian e-commerce dataset through speech. The initial dashboard presented five coordinated views: monthly order volume, category-level sales, state-level order distribution, review-score distribution, and delivery-time statistics. The exploration was open-ended; participants were free to pursue questions, comparisons, explanations, and data subsets that they considered relevant. They were not introduced to the three revision constructs and were not required to interrupt the assistant or follow a prescribed analytical path.

We recorded timestamped user and assistant utterances together with available interaction and dashboard events. After the exploration, participants completed a post-use questionnaire concerning when and why they changed direction, what state should be preserved or abandoned, their perceived control, and the dashboard operations they considered important.

Our analysis followed a hybrid deductive-inductive process. We first used an LLM-assisted pass to retrieve candidate episodes from the logs. Each candidate included the exact utterance, its preceding analytical commitment, proposed labels, timing, and a rationale. We then manually reviewed every candidate against the surrounding dialogue and dashboard context.

An episode was retained as an *analytical revision* only when an active analytical commitment was identifiable before the utterance and the utterance materially changed, qualified, or superseded it. Initial formation of a question or explanation was distinguished from revision of an existing one. Likewise, a change in chart, metric, or calculation procedure was not treated as a goal, hypothesis, or scope revision unless the corresponding analytical commitment also changed.

The final coding retained 27 analytical revision episodes. Using a multi-label scheme, 12 episodes involved an Analytical Goal Shift, five involved a Working-Hypothesis Revision, and 18 involved an Analytical Scope Refinement. Eight episodes involved more than one construct. Because the

categories were not mutually exclusive, their counts do not sum to the number of episodes.

Interaction timing was coded separately from revision semantics. This allowed us to distinguish *what* changed in the investigation from *when* the change was expressed relative to system speech, analytical execution, and dashboard updates.

C. Formative Findings

1) Participants revised different analytical commitments:

All four participants reported changing analytical direction during the exploration, and three reported doing so at least twice. All four reported changes to the data scope, three reported changes to their interpretations or conjectures, and two reported changes to the primary analytical question. These responses were consistent with the manually coded logs.

Goal shifts changed the knowledge outcome being pursued. Participants moved, for example, from monthly order trends to low-rating customer experience, from delivery performance to payment behavior, and from general comparison to the active search for anomalous products.

Scope refinements preserved the broader objective while changing the data target. They included geographic grouping, rating thresholds, time intervals, product-category substitution, and changes from monthly to weekly granularity. Scope changes included narrowing, broadening, substitution, and regrouping; we therefore use *Scope Refinement* rather than the narrower term *Scope Narrowing*.

Working-hypothesis revisions occurred when participants questioned or qualified an active explanation or decision criterion. They were distinguished from hypothesis formation, in which an explanation was introduced for the first time, and from method revision, in which the analyst changed a metric or calculation without necessarily changing an explanatory proposition.

2) *Revision constructs overlapped within individual utterances*: Eight episodes changed more than one analytical object. For example, one participant requested:

“Apply the same analysis to the third- through fifth-ranked states; I want to find products with relatively low ratings.”

This request simultaneously restricted the geographic scope and redirected the analytical purpose from general comparison to anomaly seeking. We therefore coded it as both Analytical Scope Refinement and Analytical Goal Shift.

Such episodes indicate that goal, hypothesis, and scope are related but distinguishable dimensions of analytical commitment. They should not be implemented as mutually exclusive runtime intent classes. Instead, the system must determine which prior commitments remain valid and compose the actions required by the latest request.

3) *Analytical revision was distinct from repair and method change*: Speech overlap did not necessarily indicate analytical revision. Participants also spoke during system output to correct recognition errors, ask immediate follow-up questions, stop an overlong explanation, or provide conversational acknowledgements. For example, “I mean by state—state”

TABLE I
REPRESENTATIVE EPISODES FOR THE THREE ANALYTICAL REVISION CONSTRUCTS. UTTERANCES ARE TRANSLATED FROM CHINESE.

Construct	Prior commitment	User utterance	Interpretation
Explore monthly order trends.	“Analyze customer experience—the low-rated orders, including their time, category, and delivery conditions.”	The desired outcome changes from describing order volume to investigating low-rating customer experience and its potential contributing factors.	Analytical Goal Shift Prioritize Bahia because it has the longest delivery time among high-volume states.
“BA only has the longest delivery time. Does this take sales volume into account?”	The user qualifies the active proposition by questioning whether delivery time alone is sufficient and introduces sales volume as an additional decision criterion.	Working-Hypothesis Revision Investigate customer experience using the full set of orders.	“Now filter the global review scores to one and two.”
The higher-level objective remains unchanged, while the relevant population is restricted to low-rating orders for subsequent analysis.	Analytical Scope Refinement		

restored a likely misrecognized request rather than expressing a new analytical direction.

Changes in chart form or metric were similarly context-dependent. Combining delivery days and order volume into a ratio changed the analytical operationalization, but did not necessarily revise an explanation or data scope. These boundary cases required interpretation of the preceding commitment rather than classification from the latest utterance alone.

4) *Participants expected selective cancellation and preservation*: Three participants indicated that the assistant should stop its current speech when a new direction was expressed. Participants also expected useful committed state to be preserved: three selected still-relevant visualizations, two selected previously confirmed findings, and two selected prior conversational context.

By contrast, three participants indicated that unfinished explanations and unfinished analyses from the old direction should be abandoned, two selected pending visual updates, and all four selected details unrelated to the revised question. These responses suggest that redirection requires selective supersession rather than either preserving or resetting the entire analytical state.

Participants’ main concerns included speech-recognition errors, misinterpretation of analytical intent, overly long responses, and an obsolete direction continuing to affect the dashboard. All four agreed that they could change direction when needed and understood which request the dashboard represented. Three reported an overall sense of control, while one was neutral. Opinions about response length were divided, reinforcing the need for concise and interruptible responses.

D. Design Requirements

The formative findings informed four design requirements.

a) *DR1: Ground interpretation in active analytical commitments*.: The system should interpret a new utterance relative to the current analytical goal, working hypothesis, filters,

visualizations, and dialogue context. This context is necessary to determine whether the utterance continues, qualifies, or supersedes the active investigation.

b) *DR2: Support compound revisions through composable actions*.: Because one utterance may revise multiple analytical objects, the system should support composable, schema-grounded action sequences rather than map every utterance to one operation or one exclusive revision type.

c) *DR3: Distinguish analytical revision from repair and method change*.: Recognition corrections, clarifications, evidence requests, initial hypotheses, and changes in chart or metric should not automatically invalidate the current analytical direction. The system should preserve valid context when the user is repairing expression rather than revising analysis.

d) *DR4: Coordinate supersession across speech, execution, and visual state*.: When a new request supersedes an active response, cancellation should extend beyond audio playback. Response-dependent tool calls, query execution, and pending dashboard updates should be invalidated so that obsolete results cannot later modify the visible state. Replanning should begin from the latest committed dashboard state while preserving still-relevant filters, visualizations, findings, and dialogue context.

These requirements directly informed VerbalVis: compact dashboard-state grounding supports DR1; composable visualization tools support DR2; separate handling of repair and analytical redirection supports DR3; and response-bound execution epochs with stale-result rejection support DR4.

E. Scope of Evidence

This formative inquiry provides design-oriented evidence rather than validation of a universal taxonomy. Four participants cannot establish theoretical saturation, estimate population-level prevalence, or demonstrate that the three constructs cover every form of analytical change.

We therefore treat Analytical Goal Shift, Working-Hypothesis Revision, and Analytical Scope Refinement as

[Figure placeholder] System architecture: Frontend (Vue/Vega-Lite) ↔ Backend (FastAPI relay + DuckDB) ↔ OpenAI Realtime (gpt-realtime-2). Realtime Layer (audio streaming, VAD, barge-in, response cancel) on the left; Analytics Layer (tool calls, DuckDB, dashboard update) on the right; shared dashboard state in the middle.

Fig. 2. VerbalVis architecture. The Realtime Layer handles conversational coordination; the Analytics Layer executes tool calls asynchronously and updates the dashboard state, which is reinjected as system context after every tool call.

theory-informed, non-exhaustive, and potentially overlapping analytical lenses. The 27 verified episodes characterize the present formative corpus and inform the design of VerbalVis. The subsequent user study evaluates the resulting full-duplex interaction and orchestration mechanisms rather than testing the three constructs as a universal classification.

IV. SYSTEM DESIGN AND IMPLEMENTATION

A. Design Goals

The framework yields three design goals:

- **DG1: Continuous intent revision.** Support revision at any moment, not only at turn boundaries.
- **DG2: Rapid replanning after supersession.** Once an interruption is detected during assistant output, the system must prevent stale response work from committing, interpret the redirecting utterance, and re-plan against current dashboard state with minimal added latency.
- **DG3: Dashboard-intent synchrony.** The shared dashboard must always reflect the latest applied operations and must serve as the canonical source of truth for the LLM, so that explanations are grounded in what the user sees rather than in the model’s prior outputs.

B. Architecture Overview

VerbalVis comprises a Vue 3 + Vega-Lite frontend, a FastAPI backend with an in-memory DuckDB analytics layer, and a relay that bridges a browser WebSocket to the OpenAI Realtime WebSocket (Fig. ??). The backend exposes a single `/ws` endpoint; on connection, it constructs a `RealtimeSession` that runs two concurrent async loops — one forwarding client audio frames to OpenAI, one demultiplexing OpenAI events to the client and dispatching tool calls.

Realtime layer. `gpt-realtime-2` streams 24 kHz PCM audio in both directions. Depending on the experimental condition, the frontend uses local voice gating or push-to-talk, while the Realtime session may also expose server-side speech events such as `input_audio_buffer.speech_started` and `...speech_stopped`. On interruption-relevant events, the backend increments a *turn epoch*, records the in-flight `response_id` as invalidated, and marks tool tasks from the cancelled response as stale. In push-to-talk mode the backend additionally sends `response.cancel`; in server-interruption mode, it relies on the Realtime session’s

native interruption behavior and applies the same stale-result discipline locally.

Analytics layer. Tool calls (`response.function_call_arguments.done`) are dispatched to handlers backed by DuckDB queries. DuckDB holds two derived fact tables built from seven Olist CSVs at startup: `fact_order` (one row per delivered order, carrying order-grain attributes including `order_month`, `review_score`, `customer_state`, `delivery_days`, and per-order payment) and `fact_item` (one row per order item, additionally carrying `product_category` and per-item revenue price + freight). The two-table design lets revenue retain the correct semantics under both order-grain and item-grain aggregations; cross-grain filters on `product_category` from the order table are rewritten as `order_id IN (SELECT ...FROM fact_item WHERE ...)` subqueries by a shared `build_where` helper.

VerbalVis employs real-time full-duplex audio interaction for conversational coordination, while analytical operations are executed asynchronously through tool calls that update the dashboard state.

C. Tool Taxonomy

Following the Brehmer–Munzner multi-level typology [?] and Yi et al.’s how-axis [?], we organize VerbalVis operations along three dimensions: *why* (analytical goal), *how* (interaction primitive), and *what* (target of the operation: data, view, layout, insight). Table ?? lists a 15-operation taxonomy for the design space. The current prototype realizes five dashboard operations end-to-end as schema-based tools: `filter_data`, `remove_filter`, `highlight_visual`, `append_visual`, and `delete_visual`. This subset is deliberately narrow: it covers the operations needed for the running example and user-study tasks while keeping tool selection predictable under realtime interruption.

The implemented tools are deliberately minimal. `filter_data` centralizes global dataset narrowing (with a `field=null` convention for clearing, an `append` flag for AND-accumulation, and operator support for `eq`, `neq`, `in`, `gte`, `lte`, `between`); `remove_filter` removes one field constraint without disturbing the rest; `highlight_visual` redirects attention without changing data; `append_visual` creates a new chart at item or order grain based on whether `product_category` appears in the encoding; and `delete_visual` removes views from the shared workspace. This minimality is intentional: most analytical state changes in our scenarios can be expressed by composing these operations, which keeps the model’s tool-selection decision tree shallow and reduces the surface area on which the LLM can hallucinate.

D. Layered Prompt Design

The system prompt is constructed in four labeled sections, in line with the prompt-structure recommendations of the GPT-realtime-2 prompting guide [?]. *Identity* fixes the assistant

TABLE II

VERBALVIS TOOL TAXONOMY. FIVE OPERATIONS ARE IMPLEMENTED END-TO-END AS SCHEMA-BASED TOOLS (MARKED ✓); THE REMAINDER DEFINE THE BROADER DESIGN SPACE. *Why* FOLLOWS BREHMER–MUNZNER; *How* FOLLOWS YI ET AL.; *What* IDENTIFIES THE OPERATION TARGET.

Tool	Why	Why-Detail	How	What	Impl.
filter_data	Discover	Locate	Filter	Data	✓
remove_filter	Discover	Locate	Filter	Data	✓
aggregate_data	Discover	Summarize	Abstract/Elaborate	Data	
compare_data	Discover	Compare	Connect	Data	
change_chart_type	Discover	Explore	Encode	View	
change_encoding	Present	—	Encode	View	
change_sort	Discover	Compare	Reconfigure	View	
highlight_visual	Discover	Locate	Select	View	✓
focus_visual	Discover	Explore	Select+Reconfigure	View	
append_visual	Discover	Explore	Encode	View	✓
move_visual	Present	—	Reconfigure	Layout	
delete_visual	Discover	Identify	Reconfigure	Layout	✓
replace_visual	Discover	Explore	Reconfigure	Layout	
explain_visual	Discover	Identify	Connect	Insight	
summarize_dashboard	Discover	Summarize	Abstract/Elaborate	Insight	

as a collaborative data analyst, instructs language mirroring, and prescribes the opening behavior (greet, introduce the Olist dataset, walk through the four base views, then invite the user to choose a direction — explicitly without preemptively highlighting any view). *Dashboard Knowledge* provides a bidirectional view-id $\leftrightarrow$ Chinese-and-English-label dictionary, the exhaustive field list (forbidding fabricated fields such as “return_rate” or “profit”), and value semantics (e.g., review_score 1–2 negative; customer_state two-letter codes; revenue in BRL pronounced “reais”). *Tool Usage Rules* specify operator semantics, the cross-grain filter convention, and tool-selection guidelines that explicitly forbid defaulting to the review view for generic prompts such as “what is interesting in the data.” *Realtime Rules* legitimize interruption as a high-priority signal and instruct the model to abandon prior explanations and re-evaluate against the latest user utterance.

E. Dashboard State and Context Reinjection

The four base views (view-trend, view-review, view-map, view-category) and any user-added workspace-N charts are maintained as a single backend list. After every tool call, the backend (i) re-queries all views against the current filter set, (ii) recomputes per-view summary statistics (e.g., peak month, low-score ratio, top state, top category), (iii) pushes a views_update event to the frontend for re-render, and (iv) injects a compact textual snapshot of the dashboard — highlighted view, active filters, total filtered row count, and per-view summary — back into the LLM as a system message. This *context reinjection* pattern follows the guidance in the GPT-realtime-2 long-context section [?]: the model is never asked to reconstruct dashboard state from prior turns, only to act on what it is given.

F. Supersession Control Protocol

VerbalVis treats barge-in as a control event, not as a semantic classifier. The reusable protocol has five stages.

First, the runtime detects possible supersession through either a server speech-start event or a client push-to-talk start during assistant output. Second, the backend increments `turn_epoch`, so newly arriving user input and later tool calls belong to a new response epoch. Third, the system attempts to stop obsolete output through `response.cancel` or `conversation.item.truncate`, depending on the input mode and Realtime event sequence. Fourth, tool calls associated with the superseded response are cancelled when possible and otherwise checked with `is_stale_tool_call` before any result is returned to the model or applied to the dashboard. Fifth, VerbalVis reinjects a compact dashboard snapshot before the next `response.create`, so the revised plan is grounded in the latest valid visual state.

When barge-in is enabled, an interruption is represented by either a server speech-start event or a client push-to-talk start during assistant output. The backend then:

- 1) Increment the per-session `turn_epoch`; record the in-flight `response_id` as invalidated.
- 2) Cancel all in-flight tool tasks bound to the cancelled response.
- 3) Send `response.cancel` or `conversation.item.truncate` when the local input mode requires explicit cancellation or audio truncation.
- 4) Forward `speech_started` to the frontend so the UI can stop playback immediately.

A double-lock discipline (`openai_send_lock` for outbound messages, `tool_state_lock` for state mutation) and a per-response pending-tool counter ensure that (i) tool results from cancelled responses are discarded by an `is_stale_tool_call` check before being forwarded to the model, and (ii) when a single model turn issues multiple tool calls, only the final completing call triggers a fresh `response.create`, preventing duplicate or racing responses.

No-barge ablation. We define **VerbalVis-NoBarge** as the controlled ablation of the protocol. It uses the same

`gpt-realtime-2` model, voice, prompt, tool schema, dashboard, backend, VAD settings, and context-reinjection logic as `VerbalVis-FullDuplex`. The only intended difference is policy: user speech during assistant output is recorded or queued, but it does not trigger `response.cancel`, `conversation.item.truncate`, epoch invalidation, or stale-tool discard until the current response completes. This condition separates the effect of the realtime model family from the specific effect of the supersession path.

V. CASE STUDY: THREE FORMS OF ANALYTICAL INTENT REVISION

We illustrate the framework with a single session on the Olist Brazilian e-commerce dataset (96,478 delivered orders; 110,196 order items spanning 73 product categories and 27 states; September 2016 – August 2018). The analyst, a domain non-expert, was asked to explore customer satisfaction with no predetermined goal.

A. Initial State

The system greets the user, names the dataset, and walks through the four base views: monthly order trend, review-score distribution, orders by state, top-15 categories by revenue. The system then asks the analyst what they would like to explore. The analyst responds, “Let’s look at the trend.”

B. Type 1 — Goal Shift

System (mid-utterance): “Order volume peaks in November 2017, roughly 4× the average month. Let me decompose that by product category to see whether it’s a Black Friday effect across categories or concentrated in — ” *Analyst (barge-in)*: “I care more about ratings, not sales.”

The `barge-in` fires `response.cancel`; the redirecting utterance is interpreted; the controller emits `highlight_visual(view=review)` and the assistant continues: “The review distribution is heavily concentrated at 5 stars but with a non-trivial 1- and 2-star tail. Want me to filter to low-rated orders and see what they share?” The intent has shifted from a sales decomposition to a satisfaction investigation; the analyst was spared the remainder of the category breakdown.

C. Type 2 — Hypothesis Correction

System (mid-utterance): “Electronics-adjacent categories such as `computers_accessories` and `electronics` show below-average ratings. This is often attributable to product quality issues with — ” *Analyst (barge-in)*: “It’s logistics, not the products.”

The system invokes `append_visual(scatter, delivery_days, review_score)` followed by `filter_data(review_score, lte, 2)`; the resulting scatter shows the low-rated tail clustering at high delivery days. The assistant continues: “You’re right — low-rated orders disproportionately have delivery times above 20 days; the product-quality story doesn’t hold up.” The intent revision substituted a delivery hypothesis for a product-quality hypothesis, and the assistant aborted its prior explanation rather than completing it.

D. Type 3 — Scope Narrowing

System (mid-utterance): “Among states with the worst delivery times, the northeast — Bahia and Ceará — have averages above 25 days, while — ” *Analyst (barge-in)*: “Just São Paulo.”

`filter_data(customer_state, eq, "SP")` is issued; all four base views recompute against the SP-only subset; the assistant pivots to “Within SP, average delivery is 8.4 days; the low-rated tail is much smaller than the national figure.” Scope contracted from a national survey to a state-level focus, exploiting the analyst’s prior interest in SP without the analyst having had to articulate that interest at the start of the session.

E. Summary

Each revision corresponds to one of the three framework types; the dashboard, the system narration, and the model’s working hypothesis remained synchronized throughout because every superseded response was followed by a fresh `conversation.item.create` of dashboard context before the next `response.create`. We argue this synchrony is the practical value of treating barge-in as a structural supersession signal rather than merely as audio cancellation.

VI. USER STUDY AND TECHNICAL BENCHMARK

A. Design

The study is designed to test the paper’s central claim rather than general usability: whether a full-duplex supersession pathway makes analytical intent revision in EDA faster, more natural, and more controllable. The primary causal comparison is **not** between a realtime speech model and a cascaded speech pipeline. Instead, we compare two conditions that share the same model, voice, prompt, tool schema, dashboard, backend, VAD configuration, and context-reinjection logic. The only intended difference is whether user speech during assistant output triggers immediate response cancellation, epoch invalidation, and stale-action discard.

We plan a within-subjects, counterbalanced study with $N=12-16$ participants. Participants must have basic data-analysis experience, but need not know the Olist domain in advance. Each participant completes two open-ended exploration tasks on the Olist dataset, one per primary condition; task-condition assignment and condition order are counterbalanced. Each task lasts 12–15 minutes after a short tutorial.

- **Condition A (VerbalVis-FullDuplex)**: audio is handled by the `gpt-realtime-2` session; the user can speak during assistant output; barge-in cancels or truncates the current response, increments `turn_epoch`, invalidates stale tool work, and triggers replanning against reinjected dashboard context.
- **Condition B (VerbalVis-NoBarge)**: the system uses the same `gpt-realtime-2` session, voice, prompt, VAD settings, dashboard, backend, tool schema, and context-injection logic. However, user speech during assistant output is queued until the current response completes; it does not trigger `response.cancel`, `conversation.item.truncate`, immediate epoch

invalidation, or stale-tool discard. This preserves the realtime model family while removing the supersession policy.

Supplementary ecological baseline. We additionally define **VerbalVis-Cascade** as an ecological baseline rather than the main causal comparison. In this condition, user speech is transcribed by ASR, passed to a text LLM using the same dashboard tool schema and Olist backend, and synthesized back to speech with TTS. The user must wait for the system to finish before the next request is processed. This baseline represents the speech-wrapped, turn-based text-LLM VA regime implicit in systems such as LightVA and DashChat, but it necessarily changes model type, speech pipeline, ASR/TTS latency, audio quality, turn-taking policy, and response timing. We therefore report Cascade results as supplementary context, not as evidence that the barge-in mechanism alone caused any observed advantage.

B. Tasks

Tasks are written to invite open-ended EDA while naturally eliciting the three revision types in our framework. Participants are not instructed to interrupt. The tutorial only explains the interaction regime of each condition: in FullDuplex they may speak whenever they want to redirect the analysis; in NoBarge their speech during assistant output is accepted but processed after the current response finishes. If Cascade is run, participants are told that it is a turn-based speech interface whose next request is processed only after the current response completes.

- **Customer satisfaction task.** “Explore what factors may be associated with low customer review scores. Produce three findings and one follow-up recommendation.”
- **Sales, logistics, and region task.** “Investigate unusual order or revenue patterns and whether they relate to delivery or regional differences. Produce three findings and one follow-up recommendation.”

C. Measurements

Quantitative.

- *Intent revision count* — number of user-initiated direction changes per session, coded from session video, transcript, dashboard events, and the `barge_in.log` timeline.
- *Revision type distribution* — proportion of revisions coded as Goal Shift, Hypothesis Correction, or Scope Narrowing.
- *Exploration breadth* — number of distinct analytical dimensions touched (fields, views, filter expressions) during the session.
- *Insight count* — number of substantive findings reported in the post-task summary, coded by two raters with Cohen’s κ reported.
- *Time-to-revision* — elapsed time between the start of the redirecting utterance and the first system action aligned with the new intent, measured from video and event logs.
- *Wasted output time* — time during which the system continues producing the old analytical direction after the participant has begun redirecting.

- *Queue delay* — in NoBarge, the interval between redirecting speech onset and the moment the queued request begins processing.
- *Time-to-first-audio* (*TTFA*) and *tool-call duration*, recorded automatically by the existing `_response_metrics` instrumentation in `realtime.py`; if Cascade is run, ASR, text-LLM, and TTS latencies are logged separately and reported as component-level ecological costs.

Qualitative.

- Think-aloud transcripts, coded thematically.
- Semi-structured post-task interviews (10–15 min).
- NASA-TLX (six standard sub-scales).
- Perceived Control scale (3 Likert-7 items, e.g., “I felt able to redirect the analysis whenever I wanted.”).

D. Episode Coding

The unit of analysis is a *revision episode*. Two coders inspect synchronized transcripts, screen recordings, dashboard event logs, and audio timelines. For each candidate episode, coders determine (i) whether it is a valid analytical intent revision rather than a clarification, backchannel, or unrelated speech fragment; (ii) whether it is Goal Shift, Hypothesis Correction, or Scope Narrowing; (iii) whether the system successfully responded to the revised intent; and (iv) the time from the start of the redirecting utterance to the first aligned system action, such as a tool call, dashboard update, or explicit spoken acknowledgment. Disagreements are resolved through discussion after reporting inter-rater agreement.

E. Hypotheses

- **H1:** Intent revision count is higher under VerbalVis-FullDuplex than under VerbalVis-NoBarge.
- **H2:** VerbalVis-FullDuplex reduces time-to-revision, queue delay, and wasted output time relative to VerbalVis-NoBarge.
- **H3:** VerbalVis-FullDuplex increases exploration breadth and insight count relative to VerbalVis-NoBarge.
- **H4:** Perceived control is higher under VerbalVis-FullDuplex, while NASA-TLX is not significantly higher.
- **H5:** The advantage of VerbalVis-FullDuplex is largest for *Hypothesis Correction* and *Scope Narrowing*, because these revisions are most costly when users must wait through an explanation they already intend to reject or constrain.

F. Technical Benchmark

To separate mechanism-level behavior from participant variability, we additionally plan a 250-trial automated benchmark comparing VerbalVis-FullDuplex and VerbalVis-NoBarge. Scripted audio scenarios cover five interruption families: no interruption, early goal shift, late goal shift, hypothesis correction, and scope narrowing. Each scenario is repeated across multiple seeds and dashboard states. The benchmark reports speech-onset detection latency, cancel/truncation latency, stale-tool discard rate, queued-request delay, post-supersession tool

correctness, dashboard-context consistency, and component-level latency. The user study tests whether supersession-enabled full-duplex interaction changes analytical behavior and perceived control; the technical benchmark tests whether the mechanism itself reliably cancels stale work, discards stale calls, and preserves dashboard-context consistency.

G. Results (to be reported)

Placeholder. Tables and figures will report per-condition means and within-subject differences for the primary FullDuplex–NoBarge comparison with Wilcoxon signed-rank tests and effect sizes. Cascade, if collected, will be reported in a separate supplementary table to contextualize ecological latency and user experience, not to support the causal barge-in claim. Qualitative themes will be illustrated with representative quotations. The instrumentation already in place yields TTFA, per-turn latency, tool-call duration, stale-call status, and a per-session dashboard context snapshot for every tool invocation, providing both quantitative and trace data.

H. Anticipated Discussion

We expect the analyses to show that (a) the three revision types in our framework appear in the data with distinguishable distributions across participants; (b) the FullDuplex advantage over NoBarge is largest on Hypothesis Correction and Scope Narrowing episodes, where delayed processing forces users to passively witness explanations they already intend to reject or constrain; and (c) Perceived Control correlates positively with intent revision count and negatively with wasted output time. We further expect the automated benchmark to show that the gain is not merely due to the realtime model family, but to the supersession pathway that cancels stale output and preserves dashboard-context consistency.

VII. DISCUSSION

Limitations. The current prototype implements five of the fifteen taxonomy operations as schema-based tools; while these operations cover the running example and planned study tasks, claims about the taxonomy’s completeness require broader deployment. VerbalVis treats all barge-in events with the same downstream processing and does not classify revision type at runtime; the three types in Section ?? are analytical categories, not separate system policies. The planned study size ($N=12-16$) is appropriate for an exploratory within-subjects system study but still limits statistical power for subgroup analysis. VerbalVis-NoBarge is a controlled ablation rather than a naturally deployed product: it preserves the realtime model family while disabling immediate supersession, which is useful for causal attribution but less ecologically familiar. Conversely, VerbalVis-Cascade is ecologically meaningful because it represents a speech-wrapped text-LLM VA regime, but it changes model type, speech pipeline, latency decomposition, audio quality, and turn-taking policy; we therefore treat it only as supplementary context. Finally, the Olist dataset, while rich, is domain-specific; generalization to clinical, financial, or scientific datasets remains future work.

Implications. Treating barge-in as a structural supersession signal expands the conversational VA design space without requiring the system to infer the user’s meaning from timing alone. The interruption event gives the controller permission to protect the analytical state from stale work; the redirecting utterance provides the content of the revised intent. Future systems may also use the *timing* of an interruption (early vs. late in a response) as a weak cue about whether the user rejects the direction, the evidence, or the conclusion, but such use should remain subordinate to the utterance content and dashboard context.

Generalization beyond voice. The framework is voice-motivated but not voice-bound. Any modality with a true interruption primitive — including streaming-LLM text chat with mid-response edits, or gestural systems that can preempt animations — can be analyzed through the same Intent Evolution lens. We see VerbalVis as a concrete instantiation of a broader design pattern.

Future work. Beyond completing the tool taxonomy and broadening data domains, we plan (i) type-aware interruption handling that differentiates Goal Shift, Hypothesis Correction, and Scope Narrowing at runtime, (ii) a longitudinal study examining whether users’ intent-revision strategies stabilize over repeated sessions, and (iii) an extension to multi-analyst sessions where barge-ins from different participants might conflict or require negotiated replanning.

VIII. CONCLUSION

We have presented VerbalVis, a full-duplex conversational visual analytics system motivated by the observation that user barge-in can serve as a structural supersession signal during analytical work. The Intent Evolution framework identifies three recurring revision types — Goal Shift, Hypothesis Correction, and Scope Narrowing — and motivates a reusable supersession control protocol for conversational EDA. Built on `gpt-realtime-2`, response epochs, schema-based tool calling, and an in-memory DuckDB analytics layer with continuously reinjected dashboard context, VerbalVis demonstrates how full-duplex speech can move intent revision from a between-turn afterthought to a continuously available action. A case study on the Olist e-commerce dataset illustrates all three revision types in a single session; a planned within-subjects user study will make its primary comparison between VerbalVis-FullDuplex and VerbalVis-NoBarge, with VerbalVis-Cascade retained only as a supplementary ecological baseline; and a 250-trial technical benchmark will test the reliability of the supersession mechanism. Together, these evaluations separate the human effect of full-duplex redirection from the mechanism-level question of whether stale work can be canceled or discarded while dashboard context remains consistent.